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RESEARCH

Purely Periodic, Second-Order Terms of the J_2-Problem in Closed Form and Arbitrary Variables

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Abstract
For the J_2-problem, the mean-to-osculating transformation that guarantees the exact separation of long- and short-period terms up to the second order of J_2 has been recently reported by the author. The transformation was computed in Delaunay canonical variables. However, because such kind of solution is derived from a vectorial generating function, it can be obtained in arbitrary variables—either singular or not, canonical or non-canonical—without need of recomputing the perturbation solution. The procedure is illustrated for the insightful set of semi-equinoctial variables. It could be equally applied, if desired, to compute a second-order, closed-form, analytical model truly consistent with the Draper semianalytical satellite theory.

Keywords Perturbed Kepler motion · Semianalytical methods · Mean elements · Lie transforms · Noncanonical perturbation theory · Hamiltonian perturbations

1 Introduction

In the theory of artificial satellites, mean variations in strict agreement with the average behavior of osculating elements are routinely obtained with the generalized method of averaging [1–3]. To that aim the original, osculating, variation equations of the perturbation problem are formulated in a given set of variables, which are customarily chosen nonsingular. The equations of the mean-element rates, as well as the short-period corrections that provide the transformation from mean elements to osculating elements, are then constructed in the same set of variables according to the perturbation method. The mean elements are efficiently integrated with numerical methods, whereas the osculating elements are recovered from them using analytical corrections. This semianalytic integration scheme based on the decomposition of the motion into long- and short-period effects gives rise to very efficient orbit

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propagation tools. Among them, the Draper semianalytical satellite theory (DSST in what follows) stands out due to its sustained development for the last fifty years [4, 5].

Still, the generalized method of averaging is not the sole option for materializing the semianalytical framework. The same results can be obtained with the method of Lie transforms [6–8], which in addition enjoys greater versatility. Indeed, the fact that the short-period corrections are derived from a generating function, which is of vectorial character in the more general approach, makes to construct the semianalytical solution in different ways possible when using the method of Lie transforms.

Such versatility of the method of Lie transforms is certainly useful in the decomposition of the motion into the mean-to-osculating transformation and the long-period variations. The exact separation of the perturbed dynamics into short- and long-period terms is not really a constraint of the semianalytic approach, although it may be required in specific applications dealing only with the mean-element dynamics. On the contrary, since mean elements can be defined in a variety of ways, the decomposition of the perturbation solution into the mean-element dynamics and the mean-to-osculating transformation may adopt different formulations [9]. But an extra advantage of the method of Lie transforms stems from the fact that, once the generating function has been obtained in a desired set of variables, the mean-to-osculating transformation can be reformulated in a different set of variables without need of recomputing the whole perturbation theory from scratch [6, 8].

That the possibility of reformulating a perturbation solution in a different set of variables, instead of its complete rederivation, is a definitive advantage is clearly recognized when the mean-to-osculating transformation is approached in closed form—a common wish in artificial satellite theory due to the variety of satellite missions involving highly eccentric orbits. Indeed, because usual perturbations of the Keplerian motion do not depend explicitly on the mean anomaly, but through its implicit dependence on the true or eccentric anomalies, the analytical computation of the mean-to-osculating transformation in closed form confronts the solution of nontrivial integrals beyond first-order effects. Rules for solving the majority of these integrals are based on the geometric relations of the ellipse as well as on the dynamics of the Kepler problem. Therefore, it looks wise, as well as more expedite, to rely on previous, well-tested results in the literature based on the latter [10–17]. In spite of the fact that different sets of variables may be better suited for dealing with particular perturbations of the orbital motion than the traditional set of Keplerian elements [18], approaching from the beginning the closed-form solution of the required definite and indefinite integrals in a different set of variables could complicate unnecessarily the closed-form computation of the perturbation theory.

Of particular interest in different astrodynamical applications is the solution of the J_2-problem. Because it deals with the dominant gravitational perturbation of most circumterrestrial orbits, it was traditionally known as the main problem of artificial satellite theory [19]. Different perturbation theories succeeded in the computation of high-order effects of J_2 in closed form [16, 20, 21]. However, these kinds of solutions usually proceeded by partial Hamiltonian normalization to achieve the total Hamiltonian reduction, in which case the strict splitting of terms of different nature is irrelevant. Moreover, even if the exact separation were desired, achieving

the correct partitioning could not be possible due to the Hamiltonian nature of the employed methods [22]. A final, noncanonical tuning is needed in that case in order to obtain the exact isolation of the short-period terms that are purely periodic in the mean anomaly [23].

Therefore, applications looking for the exact separation of the mean variations beyond the first order of J_2 seem to be doomed to relying on noncanonical perturbation methods. When such kind of solution is approached with the generalized method of averaging, one remains tied to computing the mean-to-osculating transformation in the same variables in which the mean variations are evaluated.

The example provided by DSST of the inconvenience one finds in this process when the closed form is desired is quite illustrative. The difficulties seemed unsurmountable at the beginning, leading DSST developers to the alternative search of approximate solutions based on expansions of the elliptic motion [24]. Publication of an analytic procedure for computing the mean variations in closed-form was delayed by about twenty years [25]. Yet, the implementation of the closed-form solution for the mean variations was not reported until fourteen additional years [26]. Refer to the latter reference for more detailed descriptions. Still, the proposed closed-form formulation of the mean variations was found not optimum due to the fact that it needed numerical quadrature for its evaluation, and the highly-desired analytical formulation remained open. Recent patches provided both the mean-to-osculating transformation and the mean variations in closed form without resorting to numerical quadrature [27]. However, the purely periodic nature of the mean-to-osculating transformation of such solution is not guaranteed for the use of Hamiltonian methods in its derivation. This fact led subsequent attempts in modeling the second-order of J_2 terms to give it up with the closed form. Instead, the efforts were concentrated in improving the evaluation of analytical short-period corrections based on series expansions [28].

The wanted closed-form theory providing the exact separation of short- and long-period terms up to the second order of J_2 was finally announced in [29] (see, also, [30]). The approach followed there relied on the noncanonical version of the method of Lie transforms [7, 8] and the use of Delaunay canonical variables. The latter notably facilitated the closed-form solution of the quadratures and indefinite integrals emerging in the Lie transforms procedure. On the other hand, because the third component of the angular momentum is one of the Delaunay variables, the perturbation solution to the J_2-problem is additionally simplified in view of this variable is an integral of the osculating dynamics. However, as it happens with the Keplerian elements, the Delaunay variables present important drawbacks in the treatment of low-eccentricity orbits. In particular, the formal series representing the perturbation solution lack the expected convergence rate of a Taylor series expansion in powers of J_2 in the limited time interval in which the perturbation series apply. Nevertheless, as it was pointed out in [30], obtaining a solution in a different set of variables only requires a reformulation based on the vectorial generating function in Delaunay variables.

The challenge is accepted in this paper. More specifically, the closed-form perturbation theory is reformulated in the orbital set of semi-equinoctial variables, while preserving the purely periodic character of the mean-to-osculating

transformation. The proposed reformulation is new, to our knowledge, and may serve as a guide for DSST developers to obtain the long-awaited second-order analytical J_2-model in closed-form truly consistent with the Draper theory.

The semi-equinoctial variables replace the mean anomaly, the argument of the periapsis, and the eccentricity, by the mean distance to the node and the projections of the eccentricity vector in the nodal frame. To make an orbital set, they are complemented with the right ascension of the ascending node, the third component of the angular momentum vector, and the Delaunay action [31]. The customary choice of replacing the two later by the inclination and the semimajor axis is here disregarded in order to keep the reduced dimension of the differential system.

The obtained results are totally satisfactory, providing the expected accuracy of a second-order perturbation theory of this kind. The singularity of the semi-equinoctial variables for equatorial orbits, in which case the node is not defined, is not of worry. The nature of this singularity is not geometric and can be avoided by a rotation of the reference frame, or by a change to variables specifically adapted to this case [32, 33]. In the end, an orbit propagator suite commonly relies on different perturbation theories. Undeniably, reformulating the theory in Poincaré's nonsingular variables or in any of the variants of the set of equinoctial variables will solve the issue. However, this case is not tackled and the assignment is left open. Those interested, if any, should be able to replicate the steps described in this paper by themselves in order to achieve an analytical mean-to-osculating transformation that is purely periodic in the fast Poincaré or equinoctial variable.

The rest of the paper is organized as follows. The formulation of the J_2-problem differential system in the Delaunay variables is recalled in Sect. 2, where the variations of the semi-equinoctial elements are also displayed to check that they are free from geometric singularities. The steps needed in the reformulation of the closed-form solution of [29, 30] in the semi-equinoctial variables are detailed in Sect. 3. A way of refining the second-order solution without need of computing the third-order terms of the vectorial generator is described in Sect. 4. These higher-order refinements may be considered superfluous for a real perturbation problem due to the variety of effects that must be taken into account and are missed by the J_2-problem. However, their computation clearly illustrates the philosophy of the perturbation method as well as the fundamental need of a proper initialization of the mean elements, so many times ignored. Finally, illustrative tests of the performance of the new solution are provided in Sect. 5. For real orbits, the examples chosen would undergo clearly observable effects due to the atmospheric drag, which is not included in the model. However, this is not the point, as far as they illustrate the performance of the perturbation theory, and permitted us to compare it with previously reported tests in the literature.

For completeness, the basic equations of the Lie transforms method for vectorial flows are summarized in Appendix A. Yet, some acquaintance of the reader with the method is assumed, who is referred to the pertinent literature for details. For the same reason, needed equations from the literature are provided in Appendix B.

2 The J_2-Problem

The potential of the J_2 problem—the main problem of artificial satellite theory [19]—is

$$\mathcal{V} = -\frac{\mu}{r} + \frac{\mu}{r} J_2 \frac{R_\oplus^2}{r^2} P_2(\sin\varphi), \tag{1}$$

where r is the radius from the Earth's center of mass, P_2 is the Legendre polynomial of degree 2, and φ is geocentric latitude. The gravity field is defined by the gravitational parameter μ, the nondiemnsional oblateness coefficient J_2, and the scaling factor $R_\oplus$ for the Earth equatorial radius. Because $J_2 = \mathcal{O}(10^{-3})$ for the Earth, Eq. (1) takes the form of a perturbation of the Kepler potential given by the first summand in the right side of Eq. (1).

The flow stemming from the main problem potential can be solved in different sets of coordinates or elements. The traditional Keplerian elements $(a, e, i, \Omega, \omega, M)$ —for the semimajor axis, eccentricity, inclination, right ascension of the ascending node, argument of the perigee, and mean anomaly, respectively—are a commonly preferred set for their orbital insight. However, the equivalent set of Delaunay variables $\ell = M$, $g = \omega$, $h = \Omega$, $L = \sqrt{\mu a}$, $G = L\sqrt{1-e^2}$, and $H = G\cos I$, has the additional advantage for the main problem that the integral stemming from the axial symmetry of Eq. (1) results in the conservation of the variable H, thus reducing by one the dimension of the differential system to solve. In these variables, the variation equations take the form

$$\frac{\mathrm{d}\ell}{\mathrm{d}t} = n + \frac{\partial \mathcal{V}}{\partial L}, \tag{2}$$

$$\frac{\mathrm{d}g}{\mathrm{d}t} = \frac{\partial \mathcal{V}}{\partial G}, \tag{3}$$

$$\frac{\mathrm{d}h}{\mathrm{d}t} = \frac{\partial \mathcal{V}}{\partial H}, \tag{4}$$

$$\frac{\mathrm{d}L}{\mathrm{d}t} = -\frac{\partial \mathcal{V}}{\partial \ell}, \tag{5}$$

$$\frac{\mathrm{d}G}{\mathrm{d}t} = -\frac{\partial \mathcal{V}}{\partial g}, \tag{6}$$

$$\frac{\mathrm{d}H}{\mathrm{d}t} = 0, \tag{7}$$

where $n = \mu^2/L^3$ is the mean motion. Taking into account that $\sin\varphi = \sin i \sin\omega$, and $r = p/(1 + e\cos f)$, where f denotes the true anomaly and $p = G^2/\mu$, the explicit computation of the right sides of the variation equations is a simple exercise of

partial differentiation using the previous definition of the Delaunay variables in terms of the orbital elements. Particular arrangements useful for the perturbation approach can be found in [29, 30], for instance.

However, an inconvenience of the Delaunay variables is that the argument of the perigee is not defined in the case of circular orbits, in which $G = L$. Moreover, the eccentricity $e = (1 - G^2/L^2)^{1/2}$ appears in denominators of the variations of the mean anomaly and the argument of the perigee, Eqs. (2) and (3), respectively, which become singular in that case. Offending divisors of this kind appear also in the series defining a perturbation solution based on Delaunay variables. This fact degrades the convergence of the perturbation solution of low-eccentricity orbits, thus notably constraining its applicability. But the trouble is not related to the dynamics, it is the choice of variables, and hence it is totally avoided when using nonsingular variables.

Among the different sets of orbital elements avoiding the singularity for circular orbits [18], those based on the semi-equinoctial elements $\chi = e\cos\omega$, $\zeta = e\sin\omega$, and the mean distance to the ascending node $\lambda = M + \omega$, benefit from the insight on the long-term evolution provided by the components of the eccentricity vector in the orbital plane (χ, ζ). The variations of these elements are readily obtained from corresponding ones in Delaunay variables by standard application of the chain rule. Straightforward operations yield,

$$\frac{\mathrm{d}\chi}{\mathrm{d}t} = \frac{\chi}{e}\frac{\eta^2}{e}\left(\frac{1}{L}\frac{\mathrm{d}L}{\mathrm{d}t} - \frac{1}{G}\frac{\mathrm{d}G}{\mathrm{d}t}\right) - \zeta\frac{\mathrm{d}g}{\mathrm{d}t}, \tag{8}$$

$$\frac{\mathrm{d}z}{\mathrm{d}t} = \frac{\zeta}{e}\frac{\eta^2}{e}\left(\frac{1}{L}\frac{\mathrm{d}L}{\mathrm{d}t} - \frac{1}{G}\frac{\mathrm{d}G}{\mathrm{d}t}\right) + \chi\frac{\mathrm{d}g}{\mathrm{d}t}, \tag{9}$$

$$\frac{\mathrm{d}\lambda}{\mathrm{d}t} = \frac{\mathrm{d}\ell}{\mathrm{d}t} + \frac{\mathrm{d}g}{\mathrm{d}t}, \tag{10}$$

whose rearrangement after replacing Eqs. (2)–(7) shows that divisions by the eccentricity no longer exist. Namely,

$$\begin{aligned}\frac{\mathrm{d}\chi}{\mathrm{d}t} = n\frac{R_\oplus^2}{r^2}\frac{3J_2}{32\eta^3}\Big\{&3e^2(5s^2-4)\sin(\omega-f) + 8e(5s^2-4)\sin\omega \\ &+ 2\left[(11s^2-10)e^2 + 2(5s^2-4)\right]\sin(\omega+f) - 24ec^2\sin(\omega+2f) \\ &+ e^2(7s^2-8)\sin(\omega+3f) - e^2(13s^2-4)\sin(3\omega+f) \\ &- 8e(5s^2-1)\sin(3\omega+2f) + 2\left[e^2(2-9s^2) - 14s^2\right]\sin(3\omega+3f) \\ &- 24es^2\sin(3\omega+4f) - 5e^2s^2\sin(3\omega+5f)\Big\},\end{aligned} \tag{11}$$

$$\begin{aligned}\frac{\mathrm{d}\zeta}{\mathrm{d}t} &= n\frac{R_\oplus^2}{r^2}\frac{3J_2}{32\eta^3}\Big\{e^2(12-13s^2)\cos(\omega-f)-8e(5s^2-4)\cos\omega\\ &\quad -2\big[3e^2(3s^2-2)+2(7s^2-4)\big]\cos(\omega+f)-8e(3s^2-1)\cos(\omega+2f)\\ &\quad -5e^2s^2\cos(\omega+3f)+e^2(13s^2-4)\cos(3\omega+f)\\ &\quad +8e(5s^2-1)\cos(3\omega+2f)-2\big[e^2(2-9s^2)-14s^2\big]\cos(3\omega+3f)\\ &\quad +24es^2\cos(3\omega+4f)+5e^2s^2\cos(3\omega+5f)\Big\},\end{aligned} \tag{12}$$

$$\begin{aligned}\frac{\mathrm{d}\lambda}{\mathrm{d}t} &= n+n\frac{R_\oplus^2}{r^2}\frac{3J_2}{32\eta^3(1+\eta)}\Big\{-8(1+\eta)\big[\eta(3s^2-2)+5s^2-4\big]-2\big[5\eta^2(3s^2\\ &\quad -2)+8\eta(4s^2-3)+29s^2-22\big]e\cos f-8(3s^2-2)e^2\cos 2f-2(3s^2\\ &\quad -2)e^3\cos 3f+e^3s^2\cos(2\omega-f)+e\big[17\eta^2s^2+8\eta(4s^2-1)+11s^2-8\big]\\ &\quad \times\cos(2\omega+f)+\big[24\eta^2s^2+16\eta(4s^2-1)+8(5s^2-2)\big]\cos(2\omega+2f)\\ &\quad +e\big[13\eta^2s^2+8\eta(4s^2-1)+47s^2-8\big]\cos(2\omega+3f)+24e^2s^2\\ &\quad \times\cos(2\omega+4f)+5e^3s^2\cos(2\omega+5f)\Big\},\end{aligned} \tag{13}$$

where $\eta = G/L$, $c = H/G = \cos i$, and $s = \sin i$.

The semi-equinoctial elements need to be complemented with three additional variables to define the state of the satellite. The semimajor axis, inclination, and right ascension of the ascending node are customary choices, yet the alternative selection of the Delaunay variables L, h, and H, allows to keep the reduced dimension of the differential system [31].

3 The Second-Order Perturbation Solution. A Reformulation with Some Care

For the aim of computing a closed-form perturbation solution to the J_2-problem, whose mean-to-osculating transformation in semi-equinoctial elements is free from long-period effects, the analogous solution in Delaunay variables reported in [29] is taken as the basis. For completeness, the needed formulas of Henrard's approach to the Lie transforms method [8] are summarized in Appendix A.

To first order the solution is a mere reformulation, in which the short-period corrections of the semi-equinoctial variables are obtained from the same vectorial generator that yields a purely periodic mean-to-osculating transformation in Delaunay variables. The right sides of the osculating variations of the Delaunay variables in Eqs. (2)–(7) take the perturbative form

$$\frac{\mathrm{d}x_j}{\mathrm{d}t} = \sum_{m\geq 0}\frac{\epsilon^m}{m!}\Phi_{j,m,0}, \quad j = 1, 2, \ldots 6,$$

where $\epsilon = J_2$, the indices $1, 2, \ldots 6$, denote the Delaunay variables ℓ, g, h, L, G, H, respectively, the only non-vanishing zeroth-order term is $\Phi_{1,0,0} = n$, from which

$$\frac{\partial \Phi_{1,0,0}}{\partial x_4} \equiv \frac{\partial n}{\partial L} = -\frac{3}{a^2}. \tag{14}$$

Besides, $\Phi_{j,m,0} = 0$ for $m \geq 2$. Recall that H is an integral of the J_2-problem, and hence $\mathrm{d}x_6/\mathrm{d}t = 0$.

To avoid the introduction of secular terms in the solution of the vectorial generator, obtained from the integration of Eqs. (A15)–(A16), it is chosen

$$\Phi_{j,0,m} = \langle \tilde{\Phi}_{j,0,m} \rangle_\ell, j = 2, \ldots 5, \tag{15}$$

$$\Phi_{1,0,m} = \langle \tilde{\Phi}_{1,0,m} \rangle_\ell - \frac{3}{a^2} \langle W_{4,m} \rangle_\ell, \tag{16}$$

where $m = 1$ and $\tilde{\Phi}_{j,0,1} = \Phi_{j,1,0}$. It is readily obtained $\Phi_{2,0,1} = n(R_\oplus/p)^2 3(4 - 5s^2)/4$, $\Phi_{3,0,1} = -n(R_\oplus/p)^2(3/2)c$, $\Phi_{4,0,1} = \Phi_{5,0,1} = 0$. Next, the components $j = 2$ to 5 of the vectorial generating function $W_{j,1}$ are computed in terms of the Delaunay variables from Eqs. (A15). The indefinite integration leaves each of these components undetermined up to an arbitrary "constant," or, more properly, function of the Delaunay variables that do not depend on the mean anomaly. Therefore, Eq. (16) yields $\Phi_{1,0,1} = -n(R_\oplus/p)^2(3/4)(3s^2 - 2)\eta - 3\langle W_{4,1} \rangle_\ell / a^2$, which leaves $\Phi_{1,0,1}$ undetermined as it is $W_{4,1}$. The final indefinite integration of Eq. (A16) introduces another integration constant.

The first order terms of the mean-to-osculating transformation are then computed for the semi-equinoctial variables in terms of the Delaunay variables. Because each of them is a scalar function of the Delaunay variables, the scalar recursion in Eq. (A7) is used jointly with the scalar version of Henrard's differential operator in Eq. (A2), to obtain

$$\chi_{0,1} = \chi \frac{\eta}{e^2 L}(\eta W_{4,1} - W_{5,1}) - \zeta W_{2,1}, \tag{17}$$

$$\zeta_{0,1} = \zeta \frac{\eta}{e^2 L}(\eta W_{4,1} - W_{5,1}) + \chi W_{2,1}, \tag{18}$$

$$\lambda_{0,1} = W_{1,1} + W_{2,1}, \tag{19}$$

$$L_{0,1} = W_{4,1}, \tag{20}$$

$$h_{0,1} = W_{3,1} \tag{21}$$

Then, to get a purely periodic mean-to-osculating transformation, it is imposed that the average over the mean anomaly of these first-order terms vanish. The trivial solution $\langle W_{j,1} \rangle_\ell = 0$, $j = 1, 2, \ldots 5$, provides the needed constraints to compute the

particular values of the arbitrary constants that fully determine the generator, which are borrowed from [29] and given as Eqs. (B17)–(B21) of Appendix B. Additional, more detailed descriptions can be found in [30]. This determination of the vectorial components of the generator $W_{j,1}$ completes also the materialization of the term $\Phi_{1,0,1}$ of the mean variation of the mean anomaly.

The reformulation of the right sides of Eqs. (17)–(21) in the semi-equinoctial elements is straightforward, yielding

$$\begin{aligned}\chi_{0,1} =& \frac{R_\oplus^2}{p^2}\frac{3}{4}(5s^2-4)\zeta\phi + \frac{R_\oplus^2}{p^2}\frac{1}{32}\Big\{ -\frac{2\chi}{(1+\eta)^3}\Big[48(5s^2-4)+4(8s^2-3)(3\zeta^2\\ &-\chi^2)-4e^2\big[(3s^2-1)(3\zeta^2-\chi^2)+9(4s^2-3)\big]+4e^4(2s^2-1)+3(\eta-1)\\ &\times\big[8(5s^2-4)-2e^2(7s^2-5)+(5s^2-2)(3\zeta^2-\chi^2)\big]\Big]+6[e^2(10-11s^2)\\ &+2(4-7s^2)(\zeta^2-\chi^2)-2(5s^2-4)]\cos\theta+24(s^2+1)\chi\cos 2\theta\\ &+\big[28s^2+2e^2(9s^2-2)+(7s^2-8)(\zeta^2-\chi^2)\big]\cos 3\theta+18s^2\chi\cos 4\theta\\ &-3s^2(\zeta^2-\chi^2)\cos 5\theta+12(s^2-4)\chi\zeta\sin\theta-48(2s^2-1)\zeta\sin 2\theta\\ &-2(7s^2-8)\chi\zeta\sin 3\theta+18s^2\zeta\sin 4\theta+6s^2\chi\zeta\sin 5\theta\Big\},\end{aligned} \tag{22}$$

$$\begin{aligned}\zeta_{0,1} =& -\frac{R_\oplus^2}{p^2}\frac{3}{4}(5s^2-4)\chi\phi + \frac{R_\oplus^2}{p^2}\frac{1}{32}\Big\{ -\frac{2\zeta}{(1+\eta)^3}\Big[4e^4(4s^2-3)-4e^2\big[3(18s^2\\ &-11)+(3s^2-1)(\zeta^2-3\chi^2)\big]+48(7s^2-4)+4(8s^2-3)(\zeta^2-3\chi^2)\\ &+3(\eta-1)\big[8(7s^2-4)-2e^2(11s^2-7)+(5s^2-2)(\zeta^2-3\chi^2)\big]\Big]\\ &+12(13s^2-8)\chi\zeta\cos\theta+24(4s^2-1)\zeta\cos 2\theta+10s^2\chi\zeta\cos 3\theta\\ &-18s^2\zeta\cos 4\theta-6s^2\chi\zeta\cos 5\theta-6\big[3e^2(3s^2-2)+4(\zeta^2-\chi^2)+2(7s^2\\ &-4)\big]\sin\theta+24s^2\chi\sin 2\theta+\big[2e^2(9s^2-2)+5s^2(\zeta^2-\chi^2)+28s^2\big]\sin 3\theta\\ &+18s^2\chi\sin 4\theta-3s^2(\zeta^2-\chi^2)\sin 5\theta\Big\},\end{aligned} \tag{23}$$

$$\lambda_{0,1} = -\frac{R_\oplus^2}{p^2}\frac{3}{4}(5s^2-4)\phi + \frac{R_\oplus^2}{p^2}\frac{1}{32(1+\eta)}\Big\{\frac{4\zeta\chi}{(1+\eta)^2}\big[4\eta^3 s^2 + \eta^2(19s^2-8) - 12c^2\eta + s^2 - 4\big] + 6\big[(s^2-2)\chi^2 - (s^2+2)\zeta^2 - 2\eta(5s^2-6) - 4(6s^2-5)\big]\chi\sin\theta + 12\big[(3s^2-2)(\zeta^2-\chi^2) + (\eta+1)(5s^2-2)\big]\sin 2\theta + \big[8(6s^2-1) + 4\eta(5s^2-2) - (7s^2-4)\chi^2 - (12-17s^2)\zeta^2\big]\chi\sin 3\theta - 18s^2(\zeta^2-\chi^2)\sin 4\theta - 3s^2(3\zeta^2-\chi^2)\chi\sin 5\theta + 6\zeta\big[4(10s^2-7) + 10\eta(3s^2-2) - (7s^2-2)\chi^2 - (5s^2-2)\zeta^2\big]\cos\theta + 24(3s^2-2)\chi\zeta\cos 2\theta - \big[\zeta^2(5s^2-4) + 4\eta(5s^2-2) + (12-19s^2)\chi^2 + 8(6s^2-1)\big]\zeta\cos 3\theta - 36s^2\chi\zeta\cos 4\theta + 3s^2(\zeta^2-3\chi^2)\zeta\cos 5\theta\Big\}, \tag{24}$$

$$L_{0,1} = \frac{R_\oplus^2}{p^2}\frac{L}{32\eta^2}\Big\{-4e^2\left(\frac{2\eta^2}{1+\eta}+5\right)(3s^2-2) - 18s^2(\zeta^2-\chi^2) - 3\zeta\big[s^2(\zeta^2-3\chi^2) - (\eta^2-5)(9s^2-4)\big]\sin\theta - 24(3s^2-2)\chi\zeta\sin 2\theta + \big[2(3s^2-2)\times(\zeta^2-3\chi^2) - 9(\eta^2-5)s^2\big]\zeta\sin 3\theta + 36s^2\zeta\chi\sin 4\theta - 3s^2(\zeta^2-3\chi^2)\times\zeta\sin 5\theta + 3\big[(\eta^2-5)(3s^2-4) - s^2(3\zeta^2-\chi^2)\big]\chi\cos\theta + 12\big[(3s^2-2)\times(\zeta^2-\chi^2) - (3\eta^2-5)s^2\big]\cos 2\theta - \big[9(\eta^2-5)s^2 - 2(3s^2-2)(3\zeta^2-\chi^2)\big]\times\chi\cos 3\theta - 18s^2(\zeta^2-\chi^2)\cos 4\theta - 3s^2(3\zeta^2-\chi^2)\chi\cos 5\theta\Big\}, \tag{25}$$

$$h_{0,1} = -\frac{R_\oplus^2}{p^2}\frac{3}{2}c\phi + \frac{R_\oplus^2}{p^2}\frac{1}{4}c\Big[2\chi\zeta\frac{1+2\eta}{(1+\eta)^2} + 9\zeta\cos\theta - \zeta\cos 3\theta - 3\chi\sin\theta + \chi\sin 3\theta + 3\sin 2\theta\Big], \tag{26}$$

where $\phi = \theta - \lambda$ is the equation of the center, and $\theta = f + \omega$ is the argument of the latitude. As expected, visual inspection of Eqs. (23)–(26) confirms that the transformation is free from divisions by the eccentricity.

The argument of the latitude θ is an implicit function of the mean distance to the node λ. Its computation requires the solution of a modified form of Kepler's equation. Denoting $\upsilon = u + \omega$, where u is the eccentric anomaly, Kepler's equation $\ell = u - e\sin u$ is rewritten as $\lambda = \upsilon - e\sin(\upsilon - \omega)$, cf. [31]. Hence

$$\lambda = \upsilon + \zeta\cos\upsilon - \chi\sin\upsilon. \tag{27}$$

Once υ is solved from λ in Eq. (27), θ is computed from standard relations of the ellipse. On account of $r\cos f = a(\cos u - e)$, $r\sin f = a\eta\sin u$, straightforward operations yield

$$\frac{r}{a}\cos\theta = \cos\upsilon - \chi - \frac{\lambda-\upsilon}{1+\eta}\zeta, \tag{28}$$

$$\frac{r}{a}\sin\theta = \sin\upsilon - \zeta + \frac{\lambda-\upsilon}{1+\eta}\chi, \tag{29}$$

from which θ is determined without ambiguity.

It is worth noting that the determined integration constants yield the purely periodic character of the mean-to-osculating elements also in the Delaunay variables, so they agree with the values first computed in [29]. This is due to the linearization provided by the first-order approach. However, things are a little bit more subtle regarding the second-order terms of the mean-to-osculating transformation. While the second-order terms of the vectorial generating function $W_{j,2}$ are still those of the Delaunay variables approach, computed again from Eq. (A15)–(A16), it happens that the integration constants that make the mean-to-osculating transformation purely periodic in Delaunay variables, do not produce the same periodicity when the transformation is carried out in semi-equinoctial variables. The reason is that the combination of two purely periodic functions in some argument may bring about terms that do not average to zero in that argument, cf. [9]—think, for instance, in the cosine and cosine-square functions.

The second order of the perturbation solution in Delaunay variables starts with the computation of the tilde terms $\tilde{\Phi}_{j,0,2}$. The osculating variations only comprise terms of the zeroth- and first-order of J_2, and hence $\Phi_{j,2,0} = 0$. Then the computable terms $\tilde{\Phi}_{j,0,2} = \mathcal{L}_{j,1}(\Phi_{j,0,1}) + \mathcal{L}_{j,1}(\Phi_{j,1,0})$ are computed from the vectorial recursion in Eq. (A13) using the vectorial version of Henrard's differential operator in Eq. (A1). Next, these terms are replaced into Eqs. (15)–(16), now with $m = 2$, to compute the corresponding terms of the mean variations of the Delaunay variables. Again, the integration of the second-order components of the vectorial generator from Eqs. (A15)–(A16) introduces 5 arbitrary integration functions, which must be determined from the purely periodic character of the mean-to-osculating transformation in the semi-equinoctial elements.

Repeated iterations of the scalar version of Deprit's triangle in Eq. (A7) yield

$$x_{j,1,1} = \mathcal{L}_{0,2}(x), \tag{30}$$

$$x_{j,0,2} = \mathcal{L}_{0,1}(x_{j,0,1}) + x_{j,1,1}, \tag{31}$$

where $j = 1, 2, \ldots 5$, and each x_j is replaced by the corresponding Delaunay variable. The consequent evaluation of Henrard's scalar differential operator in Eq. (A2) results in

$$\chi_{0,2} = \sum_{l=1}^{6} \frac{\partial \chi_{0,1}}{\partial x_l} W_{l,1} + \chi \frac{\eta}{e^2 L}\left(W_{4,2}\eta - W_{5,2}\right) - \zeta W_{2,2}, \tag{32}$$

$$\zeta_{0,2} = \sum_{l=1}^{6} \frac{\partial \zeta_{0,1}}{\partial x_l} W_{l,1} + \zeta \frac{\eta}{e^2 L}\left(W_{4,2}\eta - W_{5,2}\right) + \chi W_{2,2}, \tag{33}$$

$$\lambda_{0,2} = \sum_{l=1}^{6} \frac{\partial \lambda_{0,1}}{\partial x_l} W_{l,1} + W_{1,2} + W_{2,2}, \tag{34}$$

$$L_{0,2} = \sum_{l=1}^{6} \frac{\partial L_{0,1}}{\partial x_l} W_{l,1} + W_{4,2}, \tag{35}$$

$$h_{0,2} = \sum_{l=1}^{6} \frac{\partial h_{0,1}}{\partial x_l} W_{l,1} + W_{3,2}. \tag{36}$$

The zero-average over the mean anomaly condition applied to these equations allows us to solve the terms $\langle W_{j,2} \rangle_\ell$ from a linear system, which in turn permit us to obtain the integration constants of the second-order terms of the vectorial generator, as well as to fully materialize the second-order term of the mean variation of the Delaunay variable ℓ in Eq. (16). Regarding the latter, because $\langle L_{0,2} \rangle_\ell = 0$, according to Eq. (35),

$$\langle W_{4,2} \rangle_\ell = -\sum_{l=1}^{6} \left\langle \frac{\partial L_{0,1}}{\partial x_l} W_{l,1} \right\rangle_\ell$$

which only involves the first-order components of the generator, and the first-order term $L_{0,1}$ of the transformation of the Delaunay action L. They happen to be the same as in the case in which the purely periodic character of the Delaunay variables is imposed, and hence it is so $\langle W_{4,2} \rangle_\ell$. In consequence, the second-order terms of the mean variations in Delaunay variables resulting from the choice in Eqs. (15)–(16) remain unaltered with respect to the original theory in Delaunay variables, see Eqs. (21)–(25) of [29] or (38)–(42) of [30].

The value of the integration constant obtained for $\langle W_{3,2} \rangle_\ell$ from the condition (36) also guarantees the periodicity of the mean-to-osculating transformation, but this is no longer true for the integration constant imposed by the values of $\langle W_{2,2} \rangle_\ell$ and $\langle W_{5,2} \rangle_\ell$ obtained from Eqs. (32) and (33).[1] Moreover, the square of the equation of the center resulting from the averaging of these two equations introduces nontrivial integrals in the solution of Eqs. (A15)–(A16), cf. [34]. As far as the perturbation approach is not extended beyond the second order of the mean-to-osculating transformation, discussing its properties or its efficient evaluation is declined by now, and rather Hori's advice [35] of defining a new function of the eccentricity is taken. In particular,[2]

[1] One might be tempted to simplify things and use directly the generator of the mean-to-osculating transformation of the Delaunay variables, with the hope that the long-period effects introduced in the transformation of the semi-equinoctial elements would have little effect. However, this is not at all the case due to the appearance of the eccentricity in denominators associated with these long-period terms.

[2] The solution was announced by J.F. Palacián in personal communication to the author of May 5, 2025.

$$\langle \phi^2 \rangle_\ell = \frac{1}{2\pi} \int_0^{2\pi} (f - \ell)^2 \mathrm{d}\ell = \eta \left(1 - \eta - 4 \log \frac{2\eta}{1+\eta} \right) + 2\mathrm{Li}_2 \frac{1-\eta}{1+\eta}, \quad (37)$$

where Li_2 denotes the dilogarithmic function [36, 37].

In practice, the computation of Eq. (37) can be replaced in most cases by the evaluation of a truncated power series of the eccentricity. Thus,

$$\begin{aligned} \langle \phi^2 \rangle_\ell = \Big(2 + \frac{9}{32} e^2 + \frac{43}{288} e^4 + \frac{595}{6144} e^6 + \frac{5341}{76800} e^8 + \frac{2177}{40960} e^{10} + \frac{85041}{2007040} e^{12} \\ + \frac{2556411}{73400320} e^{14} + \frac{8710273}{297271296} e^{16} + \frac{165840389}{6606028800} e^{18} \Big) e^2 + \mathcal{O}(e^{22}). \end{aligned} \quad (38)$$

The usual Horner's arrangement of this expression makes its evaluation highly efficient.

Once the new integration constants that guarantee the purely periodic character of $\chi_{0,2}$ and $\zeta_{0,2}$ have been determined, it is also needed to fix the value of the integration constant associated to $\langle W_{1,2} \rangle_\ell$ in order to guarantee also that Eq. (34) remains purely periodic in the mean anomaly.

The second-order corrections of the new, mean-to-osculating transformation are long expressions and are not printed. In these expressions the eccentricity may appear in denominators of some terms. In spite of that, there are no singularities in their evaluation. The programming issue that may happen in the evaluation of exact circular orbits is avoided by replacing them by corresponding series expansions in the eccentricity. In particular, for circular orbits it is obtained

$$\begin{aligned} \chi_{0,2}|_{e=0} = \frac{R_\oplus^4}{a^4} \frac{1}{64} \big[6(77s^4 - 200s^2 + 108) \cos\theta - (449s^4 - 346s^2 + 72) \cos 3\theta \\ + 3(17s^2 - 14)s^2 \cos 5\theta \big], \end{aligned} \quad (39)$$

$$\begin{aligned} \zeta_{0,2}|_{e=0} = \frac{R_\oplus^4}{a^4} \frac{1}{64} \big[6(155s^4 - 224s^2 + 84) \sin\theta - (395s^4 - 310s^2 + 72) \sin 3\theta \\ + 3(17s^2 - 14)s^2 \sin 5\theta \big]. \end{aligned} \quad (40)$$

In order to complete the $\mathcal{O}(J_2^2)$ theory in semi-equinoctial elements, it remains to reformulate the mean variations in these nonsingular variables too. But such reformulation only requires to plug the mean variations in Delaunay variables given in Eqs. (11)–(15) and (21)–(25) of [29]—resp. Eqs. (18)–(23) and (38)–(42) of [30]—into Eqs. (8)–(10), whose right sides are then rewritten replacing the appearances of $e = \sqrt{1 - (G/L)^2}$ and $\omega = g$ in terms of the semi-equinoctial variables χ and ζ. Namely,

$$\begin{aligned}\frac{\mathrm{d}\lambda}{\mathrm{d}t} &= n - nJ_2\frac{R_\oplus^2}{p^2}\frac{3}{4}\left[5s^2-4+(3s^2-2)\eta\right] + \frac{1}{2}J_2^2 n\frac{R_\oplus^4}{p^4}\frac{1}{\eta(1+\eta)^3}\frac{1}{512}\Big\{-96e^6\\ &\times(148s^4-29s^2-44)+e^4\left[96(887s^4-994s^2+196)-48(1003s^4-902s^2\right.\\ &\left.+28)(\zeta^2-\chi^2)\right]+e^2\left[96(2199s^4-2210s^2+192)(\zeta^2-\chi^2)-324s^4(\chi^4\right.\\ &\left.-6\zeta^2\chi^2+\zeta^4)-192(1313s^4-2164s^2+824)\right]+648s^4(\chi^4-6\zeta^2\chi^2+\zeta^4)\\ &-768(189s^4-214s^2+26)(\zeta^2-\chi^2)+192(673s^4-1052s^2+432)(1+\eta)\\ &-3(1-\eta)\Big[-2e^6(403s^4+40s^2-184)+16e^4s^2\left[3(189s^2-146)\right.\\ &\left.-(147s^2-130)(\zeta^2-\chi^2)\right]+e^2\left[16(1861s^4-1826s^2+148)(\zeta^2-\chi^2)\right.\\ &\left.-27s^4(\chi^4-6\zeta^2\chi^2+\zeta^4)-16(1953s^4-3276s^2+1216)\right]\\ &+108s^4(\chi^4-6\zeta^2\chi^2+\zeta^4)-64(383s^4-432s^2+52)(\zeta^2-\chi^2)\Big]\Big\}\end{aligned} \tag{41}$$

$$\begin{aligned}\frac{\mathrm{d}\chi}{\mathrm{d}t} &= nJ_2\frac{R_\oplus^2}{p^2}\frac{3}{4}(5s^2-4)\zeta - \frac{1}{2}J_2^2 n\frac{R_\oplus^4}{p^4}\frac{\zeta}{(1+\eta)^2}\frac{3}{64}\Big\{e^4\beta_{4,0}\\ &+e^2(\beta_{2,1}\zeta^2-\beta_{2,2}\chi^2-\beta_{2,0})+\beta_{0,0}-\beta_{0,1}\zeta^2+\beta_{0,2}\chi^2\\ &+\frac{1}{1+\eta}\left[e^4\gamma_{4,0}+e^2(\gamma_{2,1}\zeta^2-\gamma_{2,2}\chi^2-\gamma_{2,0})+\gamma_{0,1}\zeta^2+\gamma_{0,2}\chi^2\right]\Big\},\end{aligned} \tag{42}$$

$$\begin{aligned}\frac{\mathrm{d}\zeta}{\mathrm{d}t} &= -nJ_2\frac{R_\oplus^2}{p^2}\frac{3}{4}(5s^2-4)\chi + \frac{1}{2}J_2^2 n\frac{R_\oplus^4}{p^4}\frac{\chi}{(1+\eta)^2}\frac{3}{64}\Big\{e^4\beta_{4,0}\\ &+e^2(\beta_{2,2}\zeta^2-\beta_{2,1}\chi^2-\beta_{2,0})+\gamma_{0,0}-\beta_{0,2}\zeta^2+\beta_{0,1}\chi^2\\ &+\frac{1}{1+\eta}\left[e^4\gamma_{4,0}+e^2(\gamma_{2,2}\zeta^2-\gamma_{2,1}\chi^2-\gamma_{2,0})-\gamma_{0,2}\zeta^2-\gamma_{0,1}\chi^2\right]\Big\},\end{aligned} \tag{43}$$

$$\begin{aligned}\frac{\mathrm{d}h}{\mathrm{d}t} &= -nJ_2\frac{R_\oplus^2}{p^2}\frac{3}{2}c + \frac{1}{2}J_2^2 n\frac{R_\oplus^4}{p^4}c\frac{1}{(1+\eta)^2}\frac{3}{16}\Big\{e^4(5s^2+4)+e^2\left[2(15s^2-7)\right.\\ &\left.\times(\zeta^2-\chi^2)-4(42s^2-23)\right]+16(19s^2-15)-8(30s^2-13)(\zeta^2-\chi^2)\\ &-(1-\eta)\left[8(19s^2-15)-2e^2(23s^2-8)-20(7s^2-3)(\zeta^2-\chi^2)\right]\Big\},\end{aligned} \tag{44}$$

where the inclination polynomials $\beta_{i,j}$ and $\gamma_{i,j}$ in Eqs. (42) and (43) are presented in Table 1. The mean variation of L vanishes up to the second order of J_2.

Finally, the proper initialization of the mean variations from given initial conditions in osculating variables needs to know the osculating-to-mean transformation. But once the vectorial generator is computed, this transformation is obtained as a standard application of the Lie transforms theory, which only involves straightforward operations, as outlined in Appendix A.

Table 1 Inclination polynomials $\beta_{i,j}$ and $\gamma_{i,j}$ in Eqs. (42) and (43)

$\#_{i,j}$	β	γ
0,0	$8(335s^4 - 624s^2 + 288)$	$8(455s^4 - 728s^2 + 288)$
0,1	$4(525s^4 - 590s^2 + 104)$	$8s^2(25s^2 - 22)$
0,2	$4(795s^4 - 826s^2 + 104)$	$8s^2(45s^2 - 38)$
2,0	$2(845s^4 - 1132s^2 + 368)$	$4(5s^2 - 4)(79s^2 - 72)$
2,1	$2(135s^4 - 158s^2 + 28)$	$4(335s^4 - 366s^2 + 60)$
2,2	$2(195s^4 - 214s^2 + 28)$	$4(475s^4 - 486s^2 + 60)$
4,0	$45s^4 + 36s^2 - 56$	$2(225s^4 - 228s^2 + 40)$

4 Higher-Order Refinements

If progressing in closed form to higher orders in the exact separation of long-period variations and purely periodic short-period corrections, one will find the expected trouble derived from the need of using special functions in the integration of the vectorial generator. Nevertheless, this is not the case of the computation of the third-order terms of the mean variations $\Phi_{j,0,3}$, $j = 2, \ldots, 5$ in Eq. (15), which only involve partial differentiation operations and known quadratures. It is worth noting that, in spite of the terms $\Phi_{j,0,3}$ refer to the variations of the Delaunay variables, these variations are no longer the same obtained with the theory in Delaunay variables computed in [29, 30].

4.1 Third-Order Terms of the Mean Variations

The computation of the third-order term $\Phi_{1,0,3}$ of the variation of the mean anomaly from Eq. (16), requires also the knowledge of $\langle W_{4,3}\rangle_\ell$, which must be computed in such a way that it guarantees the purely periodic character of $L_{0,3}$. The latter is obtained, as before, from the scalar recursion in Eq. (A7) and the consequent use of the scalar differential operator in Eq. (A2). That is,

$$\begin{aligned} L_{2,1} &= \mathcal{L}_{0,3}(L) = W_{4,3}, \\ L_{1,2} &= \mathcal{L}_{0,2}(L_{0,1}) + \mathcal{L}_{0,1}(L_{1,1}) + L_{2,1}, \\ L_{0,3} &= \mathcal{L}_{0,1}(L_{0,2}) + L_{1,2}, \end{aligned}$$

in which $W_{4,3}$ is yet unknown, but the other terms are computable from previous orders. In particular, $L_{1,1} = \mathcal{L}_{0,2}(L) = \sum_{l=1}^{5}(\partial L/\partial x_l)W_{l,2} = W_{4,2}$ was already obtained from Eq. (30) in the computation of the second-order terms.

Then, the periodicity condition $\langle L_{0,3}\rangle_\ell = 0$ yields

$$\langle W_{4,3}\rangle_\ell = -\left\langle \mathcal{L}_{0,1}(L_{0,2}) + \mathcal{L}_{0,2}(L_{0,1}) + \mathcal{L}_{0,1}(W_{4,2})\right\rangle_\ell, \tag{45}$$

in which evaluation only known quadratures are found. In particular,

$$\begin{aligned}\langle W_{4,3}\rangle_\ell =& L\frac{R_\oplus^6}{p^6}\frac{27}{4}s^2(5s^2-4)^2(\cos 2\omega)\frac{\eta^4}{e^2}\log\frac{2\eta}{1+\eta}\\ &+L\frac{R_\oplus^6}{p^6}\frac{9}{2048(1+\eta)^4}\sum_{i=0}^{3}\sum_{j=0}^{11}Q_{i,j}(s^2)\eta^{j-3}s^{2i}\cos 2i\omega,\end{aligned} \tag{46}$$

where the inclination polynomials $Q_{i,j}$ are presented in Table 2. This value is then combined with the long-period terms stemming from the computation of $\langle\tilde{\Phi}_{1,0,3}\rangle_\ell$, to finally obtain $\Phi_{1,0,3}$ from Eq. (16).

As before, once the third-order terms of the mean variations of the Delaunay variables have been computed, the corresponding mean variations of the semi-equinoctial elements are in turn computed from Eqs. (8)–(10). Finally, the right

Table 2 Inclination polynomials $Q_{i,j}$ in Eqs. (46)

$j\backslash i$	0	3
0	$715(3s^2-2)(45s^4-24s^2+8)$	$-\frac{315}{8}$
1	$2860(3s^2-2)(45s^4-24s^2+8)$	$-\frac{315}{2}$
2	$33(13959s^6-17994s^4+9192s^2-2192)$	$-\frac{837}{8}$
3	$-4(24813s^6+10242s^4-43416s^2+14576)$	369
4	$-2(279467s^6-230170s^4+5480s^2+10928)$	$\frac{2241}{4}$
5	$-56(2741s^6-2134s^4+728s^2-176)$	-153
6	$14(24061s^6-23118s^4-200s^2+1616)$	$-\frac{2637}{4}$
7	$8(17779s^6-12858s^4-3752s^2+2000)$	-180
8	$-153927s^6+201954s^4-62664s^2+7440$	$\frac{2025}{8}$
9	$-92124s^6+118536s^4-34080s^2+4160$	$\frac{261}{2}$
10	$11859s^6-8914s^4-1912s^2+1840$	$-\frac{81}{8}$
11	$4(2391s^6-2698s^4+440s^2+80)$	-9
$j\backslash i$	1	2
0	$-\frac{15015}{8}(45s^4-48s^2+16)$	$\frac{4095}{2}(3s^2-2)$
1	$-\frac{15015}{2}(45s^4-48s^2+16)$	$8190(3s^2-2)$
2	$-\frac{33}{40}(367785s^4-381648s^2+128272)$	$\frac{9}{35}(62455s^2-37189)$
3	$\frac{3}{5}(802575s^4-917376s^2+299984)$	$-\frac{396}{35}(5185s^2-3861)$
4	$\frac{3}{20}(6353295s^4-7022832s^2+2308208)$	$\frac{2173686}{35}-\frac{614391}{7}s^2$
5	$\frac{9}{5}(35575s^4-11304s^2+2016)$	$\frac{6}{35}(174325s^2-162056)$
6	$-\frac{3309717}{4}s^4+\frac{5073164}{5}s^2-\frac{1721716}{5}$	$\frac{773232}{7}s^2-\frac{3021794}{35}$
7	$-\frac{8}{5}(294765s^4-372877s^2+132458)$	$\frac{6}{35}(120045s^2-72656)$
8	$\frac{1}{40}(7315485s^4-9121264s^2+2957456)$	$\frac{1}{70}(3155918-3679545s^2)$
9	$\frac{607149}{2}s^4-\frac{2214272}{5}s^2+\frac{843448}{5}$	$\frac{12}{35}(44963-55425s^2)$
10	$\frac{1249839}{8}s^4-261006s^2+109182$	$7671s^2-7195$
11	$37107s^4-66216s^2+28896$	$6(439s^2-414)$

sides of the mean variations are reformulated in terms of the semi-equinoctial elements.

These higher-order frequencies may help in stabilizing the growth of the secular components of the errors in the numerical propagation of the mean elements, which, soon or later, will become dominant in the perturbation solution. Moreover, they show that the mean variation of the Delaunay action—and, in consequence, that of the semi-major axis—does not vanish at third order, a fact that makes an important difference of the noncanonical approach with respect to canonical perturbation methods. In particular, $\mathrm{d}L/\mathrm{d}t = (1/3!)J_2^3\Phi_{4,0,3}$, where

$$\begin{aligned}\Phi_{4,0,3} = &- nL\frac{R_\oplus^6}{p^6}\frac{27\zeta\chi}{128\eta(1+\eta)^2}s^2(5s^2-4)\Big\{32(12s^2-5)+36s^2(\zeta^2-\chi^2)\\ &+e^2\big[4(117s^2-106)-9s^2(\zeta^2-\chi^2)\big]-16e^4(6s^2-5)\\ &+(\eta-1)\big[32e^2(11s^2-9)+18s^2(\zeta^2-\chi^2)+8(19s^2-6)\big]\Big\}.\end{aligned} \tag{47}$$

However, the perturbation theory still lacks the third-order terms of the transformation from osculating to mean variables that is needed to initialize the mean elements to the same level of accuracy. But this would need to previously solve the third-order terms of the vectorial generator, which is declined to approach.

4.2 Calibration from the Energy Integral

On the other hand, because the J_2-problem conserves the total energy, partial remedy to this inconsistency is obtained with a calibration of the initial mean semimajor axis [38]—or the equivalent Delaunay action in our particular case. To this effect the energy integral is written in the perturbative form

$$E_{\text{osculating}} = \sum_{m\geq 0}\frac{\epsilon^m}{m!}E_{m,0}, \tag{48}$$

with $\epsilon = J_2$, $E_{0,0} = -\frac{1}{2}(\mu/a)$, $E_{1,0} = (\mu/r)(R_\oplus/r)^2P_2(\sin\varphi)$, and $E_{m,0} = 0$ for $m \geq 2$. Then, the computed vectorial generating function is used to transform the energy integral into mean elements by the standard application of the scalar version of Deprit's triangle. That is,

$$E_{\text{mean}} = \sum_{m\geq 0}\frac{\epsilon^m}{m!}E_{0,m}, \tag{49}$$

where, in the mean elements,

$$E_{0,1} = \frac{\mu}{2a}\frac{R_\oplus^2}{p^2}\frac{1}{2}\eta(3s^2-2), \tag{50}$$

$$E_{0,2} = -\frac{\mu}{2a}\frac{R_\oplus^4}{p^4}\frac{3}{256\eta}\Big\{2\big[\eta^4(121s^4 - 8s^2 - 40) + 16\eta^3(3s^2 - 2)^2 - 10\eta^2(53s^4 - 8s^2 - 8) + 35(27s^4 - 24s^2 + 8)\big] + 16\Big[8(5s^2 - 4)\frac{2\eta^3 + 4\eta^2 + 6\eta + 3}{(1+\eta)^2} + \eta^2(39s^2 - 34) - 3(61s^2 - 46)\Big]s^2e^2\cos 2\omega + 9s^4e^4\cos 4\omega\Big\}. \tag{51}$$

These terms are readily written in semi-equinoctial variables noting that

$$\begin{aligned} e^2\cos 2\omega &= \chi^2 - \zeta^2, \\ e^4\cos 4\omega &= \chi^4 - 6\zeta^2\chi^2 + \zeta^4, \\ e^6\cos 6\omega &= \big(\chi^2 - \zeta^2\big)\big(\chi^4 - 14\zeta^2\chi^2 + \zeta^4\big). \end{aligned}$$

Needless to say, in spite of Eq. (48) has Hamiltonian character, Eq. (49) is no longer a Hamiltonian because the computed mean-to-osculating transformation is not canonical. So in no way the mean variations can be derived from the given integral in mean elements—save, of course, for the first-order approach, in which case the canonical and noncanonical theories match.

Undeniably, the third-order components of the generating function $W_{j,3}$ is not known, a fact that seems to prevent the computation of the third-order term of the energy integral in mean elements. However, it happens that the Keplerian term $E_{0,0} = -\frac{1}{2}(\mu/L)^2$, which only depends on L, is the only one appearing in computations involving $W_{j,3}$. Namely, from repeated application of Eq. (A7),

$$E_{0,3} = \mathcal{L}_{0,1}(E_{0,2}) + \mathcal{L}_{0,2}(E_{0,1}) + \mathcal{L}_{0,1}(E_{1,1}) + \mathcal{L}_{0,3}(E_{0,0}), \tag{52}$$

where the first three summands on the right side of Eq. (52) are computable, but the last one is not. Still, what it is certainly known is that $E_{0,3}$ must be free from short-period terms. Hence the unknown, short-period components of $\mathcal{L}_{0,3}(E_{0,0}) = nW_{4,3}$ must cancel the short-period terms resulting from the application of the other differential operators $\mathcal{L}_{0,j}$ out of Eq. (52). That is, $E_{0,3}$ must match the average of the resulting expressions on the right side of Eq. (52). Therefore, like in the previous computation of the third-order terms of the mean variations, only the average $\langle W_{4,3}\rangle_\ell$ is needed, which was already computed in Eq. (46). Therefore,

$$E_{0,3} = \frac{\mu}{2a}\frac{R_\oplus^6}{p^6}\frac{9}{32}\Big\{12(5s^2 - 4)^2\Big[2\Big(5\frac{s^2}{e^2} - 6s^2 + 1\Big)\cos 2\omega - 3s^2 + 2\Big]\log\frac{2\eta}{1+\eta} + 9(5s^2 - 4)^3\frac{e^2}{\eta^3}\langle\phi^2\rangle_\ell + \frac{1}{32}\frac{1}{(1+\eta)^4}\sum_{i=0}^{3}\sum_{j=0}^{11}P_{i,j}(s^2)\eta^{j-3}\cos 2i\omega\Big\}, \tag{53}$$

where $P_{3,j} = Q_{3,j}s^6$, $P_{i,j} = Q_{i,j}s^{2i}$, $i,j = 0,1,2$, and the remaining inclination polynomials $P_{i,j}$ are listed in Table 3. It is remarked that the evaluation of Eq. (53) is nonsingular in spite of the explicit appearance of the eccentricity as divisor. In

Table 3 Inclination polynomials $P_{i,j}$ in Eq. (53)

$j\backslash i$	0	
3	$4(13527s^6 - 93627s^4 + 106176s^2 - 30920)$	
4	$2(16953s^6 - 408270s^4 + 471000s^2 - 134704)$	
5	$16(34579s^6 - 84898s^4 + 64408s^2 - 16512)$	
6	$2(145103s^6 - 63914s^4 - 110200s^2 + 45488)$	
7	$-8(72644s^6 - 187899s^4 + 158608s^2 - 43400)$	
8	$-610543s^6 + 1130994s^4 - 717512s^2 + 172560$	
9	$-4(9897s^6 + 17978s^4 - 42376s^2 + 15376)$	
10	$142443s^6 - 327938s^4 + 267784s^2 - 76240$	
11	$12(3371s^6 - 7169s^4 + 5408s^2 - 1496)$	
$j\backslash i$	1	2
3	$457353s^6 - \frac{2580768}{5}s^4 + \frac{839472}{5}s^2$	$-\frac{409140}{7}s^6 + \frac{1523286}{35}s^4$
4	$\frac{3369465}{4}s^6 - \frac{4475844}{5}s^4 + \frac{1448916}{5}s^2$	$-\frac{599103}{7}s^6 + \frac{2109006}{35}s^4$
5	$-81789s^6 + \frac{926584}{5}s^4 - \frac{330976}{5}s^2 - 1792$	$\frac{252198}{7}s^6 - \frac{1163016}{35}s^4$
6	$-\frac{3201381}{4}s^6 + \frac{4916284}{5}s^4 - \frac{1715956}{5}s^2 + 5120$	$\frac{788184}{7}s^6 - \frac{3095994}{35}s^4$
7	$-294024s^6 + \frac{1963976}{5}s^4 - \frac{981264}{5}s^2 + 27392$	$\frac{63414}{7}s^6 - \frac{83556}{35}s^4$
8	$\frac{2334681}{8}s^6 - \frac{1697518}{5}s^4 + \frac{322962}{5}s^2 + 23552$	$-\frac{887445}{14}s^6 + \frac{1920399}{35}s^4$
9	$\frac{742893}{2}s^6 - \frac{2904672}{5}s^4 + \frac{1278328}{5}s^2 - 15616$	$-\frac{106812}{7}s^6 + \frac{429516}{35}s^4$
10	$\frac{1968207}{8}s^6 - 460734s^4 + 247550s^2 - 28672$	$14175s^6 - 13011s^4$
11	$76851s^6 - 148584s^4 + 81888s^2 - 9984$	$4050s^6 - 3798s^4$

the exact case of zero mean-eccentricity, or when terms $\mathcal{O}(e^2)$ and higher may be ignored in the series expansion of $E_{0,3}$, Eq. (53) can be replaced by

$$E_{0,3,\text{circular}} = \frac{\mu}{2a}\frac{R_\oplus^6}{a^6}\frac{27}{16}(269s^6 - 554s^4 + 408s^2 - 108),$$

thus avoiding the trouble.

Having computed the energy integral in mean elements up to the third order of J_2, one is ready for the calibration process. Thus, for given initial conditions in osculating elements the energy value $E_{\text{osculating}} = E$ is computed from Eq. (48). Next, the inverse, osculating-to-mean transformation—which is only known to the second order of J_2—is applied to compute the mean elements corresponding to these initial conditions, and replace $E_{\text{mean}} = E$ in Eq. (49), which is further rearranged as

$$E = -\frac{\mu}{2a}\left(1 + \sum_{m\geq 1}^{k}\frac{\epsilon^m}{m!}\frac{E_{0,m}}{E_{0,0}}\right) + \mathcal{O}(J_2^{k+1}), \tag{54}$$

where k denotes the truncation order at which the calibration is applied without constraint to the case $k = 3$ of current concern. If the mean value of $L = \sqrt{\mu a}$ were known to the third order, then Eq. (54) would be an identity in spite of the truncation

of the other elements to second-order effects. However, the low accuracy with which the mean value of L is known makes that the error in the computation of the energy in mean elements is in fact $\mathcal{O}(J_2^k)$.

However, if the value of the semimajor axis in Eq. (54) is replaced by the calibrated value [38]

$$a^* = \frac{\mu}{2(-E)}\left(1 + \sum_{m\geq 1}^{k} \frac{\epsilon^m}{m!}\frac{E_{0,m}}{E_{0,0}}\right), \tag{55}$$

then Eq. (54) certainly turns an identity. So the initial, mean value of L obtained from the second-order, osculating-to-mean transformation is replaced by the new one $L = \sqrt{\mu a^*}$, which would only differ from the correct value on higher-order effects.

It is remarked that the calibration process does not necessarily provides more accurate initial conditions in mean elements than those provided by the osculating-to-mean transformation. What it does provide is the initial conditions of an orbit closer to the energy manifold of the true orbit. Because of that, the improvements in the mean value of the mean motion operated by the calibration should reduce the dominant in-track errors in a long-term propagation, yet no claims are made regarding the behavior of other kinds of errors.

5 Performance Tests

To illustrate that the noncanonical perturbation solution provides the expected performance, two different examples are presented. The first one is a low-eccentricity orbit, which justifies the efforts in constructing a theory in nonsingular variables. As mentioned before, the theory based on Delaunay variables would not be adequate for propagating semi-analytically this kind of orbit due to the appearance of the eccentricity in denominators of the mean-to-osculating transformation, with the consequent lack of convergence. The other example is a Geosynchronous transfer orbit (GTO), whose eccentric is very high and is the type of orbit that justifies the efforts in obtaining the closed form of the perturbation theory. In all cases the gravitational potential is defined through the values $\mu = 398600.4415\ \mathrm{km}^3/\mathrm{s}^2$, $R_\oplus = 6378.1363$ km, $J_2 = 1.082634 \times 10^{-3}$.

The true, reference orbits used in the comparisons were computed from corresponding Runge-Kutta integrations of the original differential equations in Cartesian coordinates. To guarantee the quality of the true solutions, it has been checked that the different numerical propagations conserved the integrals of the J_2-problem—namely, the energy and the third component of the angular momentum—with at least 14 digits. Maintaining these high-accuracy standards for tests involving long-time intervals, say longer than one week, required to carry out the Runge-Kutta integrations in extended precision. On the other hand, the accuracy provided by standard numerical integration routines of popular symbolic processors was found enough to investigate the performance of the semianalytical solution in shorter time intervals.

Interested readers may consult the accompanying Wolfram Notebook,[3] where the perturbation theory is implemented, and run alternative particular examples of their choice.

5.1 Low-Eccentricity Orbit

The nominal orbital elements of the PRISMA mission [39] are chosen for the first test, which are used as osculating initial conditions of the semianalytical propagation. Namely, $a = 6878.144$ km, $e = 0.001$, $i = 97.42$, $\Omega = 168.2$, $\omega = 20$, and $M = 30$, where the angles are given in degrees. The reference orbit is computed by the numerical integration of corresponding initial conditions in Cartesian coordinates, along a time interval of two days. The orbital elements are also converted into semi-equinoctial elements $(\chi, \zeta, \lambda, L, H, h)$ and are propagated semianalytically for comparison.

To illustrate the mean features of the noncanonical, perturbation theory, four different kinds of semianalytical propagations are presented. The first one uses the first order mean-to-osculating transformation in Eqs. (23)–(26)—which are used with opposite sign to initialize the mean-element propagation—and the second-order version of the mean variations given in Eqs. (41)–(44). This theory is labeled as of the 1:2:1 type—for first-order inverse, osculating-to-mean transformation, second-order mean variations, and first-order direct, mean-to-osculating transformation. The intrinsic (along-track, radial, cross-track) errors of this propagation are presented in the left column of Fig. 1. As shown in the upper plot of that column, the improper initialization of the mean frequencies provided by this theory clearly manifests in the fast growth rate of the along-track errors, of about 1.3 km/day. The pronounced slope of the time history of the along-track errors hides the contribution of the short-period errors, whose amplitude is of about 20 meter for this type of error. Plots in the second and third rows of the left column of Fig. 1 show that the errors in the radial and cross-track directions are dominated by the amplitude of the short-period errors, which is also in the meter level due to the truncation to first order of the analytical corrections. The lower plot of this column presents the time-history of the errors of the semimajor axis, which are clearly displaced from the zero-average that should be expected from our ad hoc noncanonical approach. But this is not a defect of the theory but just another consequence of the inadequate initialization of the mean semimajor axis—or, more precisely, the Delaunay action in our chosen variables.

The flawed initialization of the semianalytical solution when using the 1:2:1 theory is mostly amended with the inclusion of second-order corrections in the osculating-to-mean transformation of the Delaunay action, which is rather approximated by a much simpler calibration of the semimajor axis based on the energy integral (the 1c:2:1-theory in our notation, the "c" standing for calibration). As shown in the upper plot of the right column of Fig. 1, now the slope of the time-history of

[3] https://notebookarchive.org/2025-05-0yxr1iu

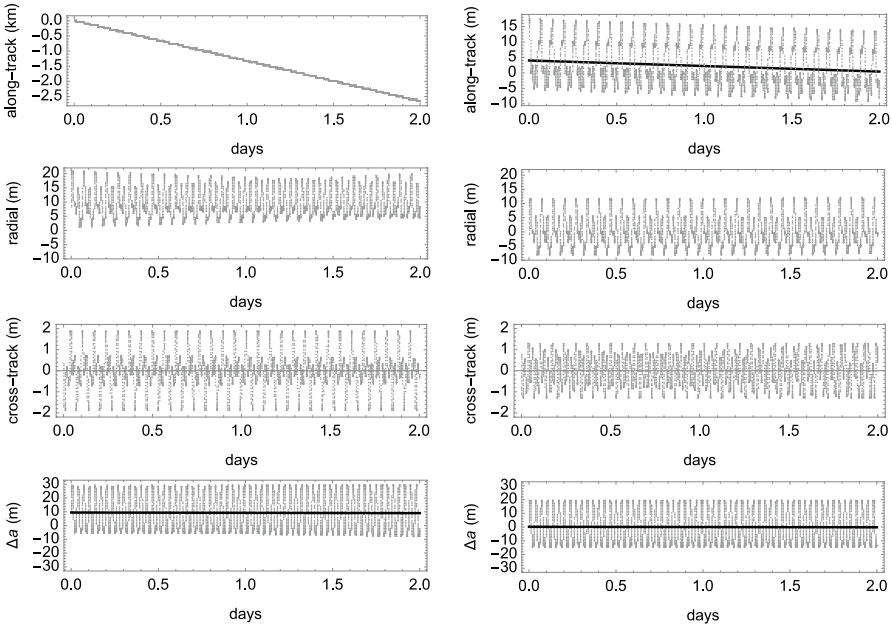

Fig. 1 Errors of a 2-day semianalityc propagation of the Prisma orbit with the 1:2:1 theory (left column) and the properly initialized version of it (right column)

the along-track errors is radically reduced to less than 2 meter per day, which is the order of magnitude expected for this type of orbit when the mean variations include second-order terms, cf. [17]. The short-period oscillations of the radial and along-track errors remain of analogous amplitude, as expected from the first-order truncation of the mean-to-osculating transformation, yet a general improvement is noticed due to the coupling of the intrinsic errors (plots in the second and third rows in the right column of Fig. 1). The lower plot of this column shows that now the errors of the semimajor axis properly average to zero, with a small shift of about 2 cm due to the unavoidable truncation of the perturbation solution.

The amplitude of the short-period components of the errors is radically reduced when the mean-to-osculating transformation takes the second order effects into account, which is called a 2:2:2 theory. Indeed, as shown in the upper plot of the left column of Fig. 2, the application of the improved mean-to-osculating transformation mostly releases the time-history of the along-track errors from short-period oscillations. The latter remain now below the meter level, as checked in the time-histories of the radial and cross-track errors shown in the second and third rows of the left column. The magnitude of the small shift from the zero-average of the errors of the semimajor axis is now clearly apparent in the lower plot in the left column of Fig. 2.

The plots in the right column of Fig. 2 show the additional improvements obtained by taking third-order terms of the mean variations into account, as well as the corresponding calibration of the mean Delaunay action with the help of the energy integral (2c:3:2 theory). The improvements of the along-track errors

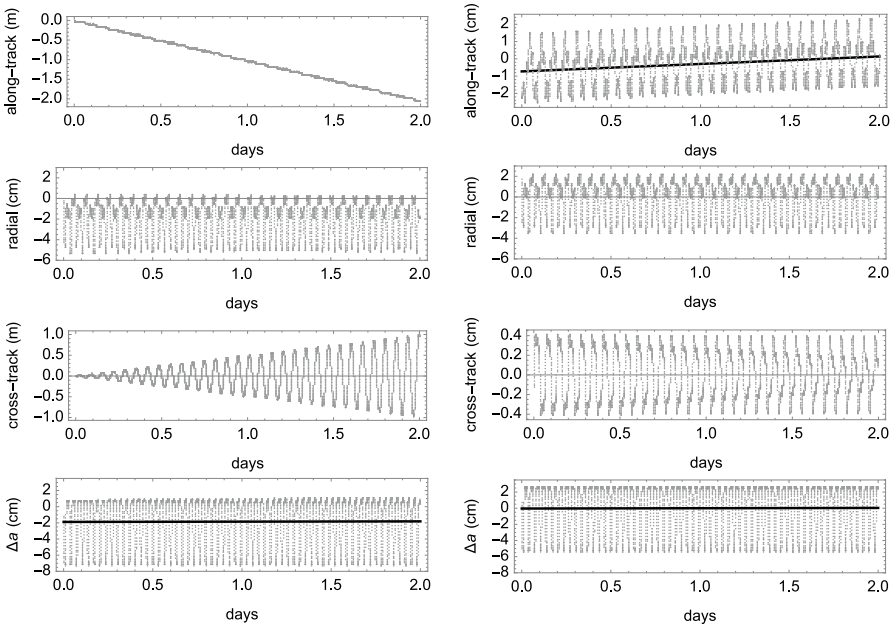

Fig. 2 2-Day semianalityc propagation of the Prisma orbit with the 2:2:2 theory (left column) and the calibrated 2c:3:2 theory (right column)

are radical, falling also to the cm level due to the amplitude of the second-order short-period corrections, and with a small secular trend of just a few mm/day (upper-right plot of Fig. 2), in contrast with the about 1 meter/day of the 2:2:2 propagation. The amplitude of the periodic components of the error is not affected, but now they show a more regular behavior with almost zero average of the radial and cross-track errors (plots in the second and third rows in the right column of Fig. 2). In agreement with the improvements of the intrinsic errors, the average of those of the semimajor axis falls to a negligible fraction of a millimeter, as depicted in the lower-right plot of Fig. 2).

Once the features introduced by the successive refinements of the noncanonical perturbation theory have been illustrated, it is worth remarking that secular errors always prevail in the long term. Therefore, depending on our particular needs, the operation of the semianalytical propagation may be notably improved by ignoring the heavy evaluation of the second-order terms of the mean-to-osculating transformation. This is illustrated in Fig. 3 for a one-month propagation of the Prisma orbit, where RSS position error denotes the root sum square of the difference between the exact position and the one obtained with the semianalytic integration, and RMS is the root mean square deviation of these errors. The figure shows that the position errors of the 2:2:2 and 1c:2:1 versions of the perturbation theory take comparable values in the long term.

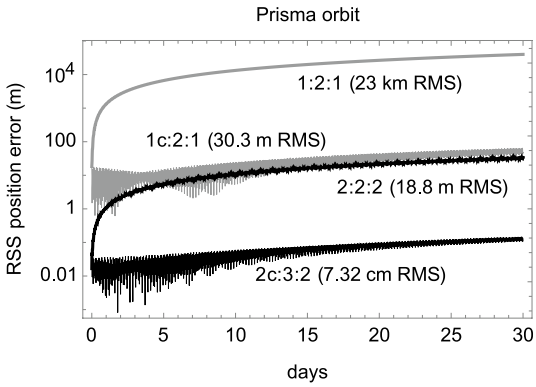

Fig. 3 Position errors of the 1-month propagation of the Prisma test orbit with the different versions of the semianalytical theory

5.2 High-Eccentricity Orbit

The second example is for a highly-eccentric orbit with nominal elements $a = 24460$ km, $e = 0.73$, $i = 30$, $\Omega = 170.1$, $\omega = 280$, and $M = 0$, with angles in degrees. As before, the reference orbit is numerically integrated for a 2-day interval from corresponding initial conditions in Cartesian coordinates, whereas the semianalytical simulations are conducted in semi-equinoctial elements $(\chi, \zeta, \lambda, L, H, h)$ from an initial state obtained from the orbital elements.

Analogous propagations to those carried out in the previous example result in the time-histories of the errors displayed in Figs. 4 and 5. The behavior of the different versions of the noncanonical semianalytical theory in the 2 days interval is essentially the same observed in the previous example, with a reduction of the along-track errors from a few km to several decameters with the straightforward calibration of the 1:2:1 theory. This accuracy is improved to the meter level when second-order effects of the mean-to-osculating transformation are taken into account, which turn into centimeters if the theory is refined with third-order terms of the mean variations and the proper initialization. The successive refinements are clearly observed too in the departure from the zero mean of the time histories of the errors of the orbital elements. In particular, for the semimajor axis, the shift of about 23 m from the zero mean resulting from the insufficient initialization of the 1:2:1 theory turns into 3.5 cm with the proper initialization of the mean semimajor axis. This offset is further reduced to about 1 cm when using the 2:2:2 theory, and to a negligible value below the mm level when the 2c:3:2 perturbation solution is used.

In spite of the much larger amplitude of the short-period errors, due to the much higher eccentricity of the tested GTO orbit, the behavior of the solution presents similar characteristics to the case of the Prisma orbit. In particular, in the 2-day interval the relative position errors remain of the order of 10^{-4} in both test cases when using the 1:2:1 version of the perturbation theory; the relative errors improve to the order of 10^{-6} in each case with the adequate initialization provided by the 1c:2:1 theory; the 2:2:2 theory reduces the relative errors to $\mathcal{O}(10^{-7})$; and the refinements provided by the third-order terms of the 2c:3:2 theory further push the relative errors to $\mathcal{O}(10^{-8})$ or below. Besides, the secular errors stemming from the truncation

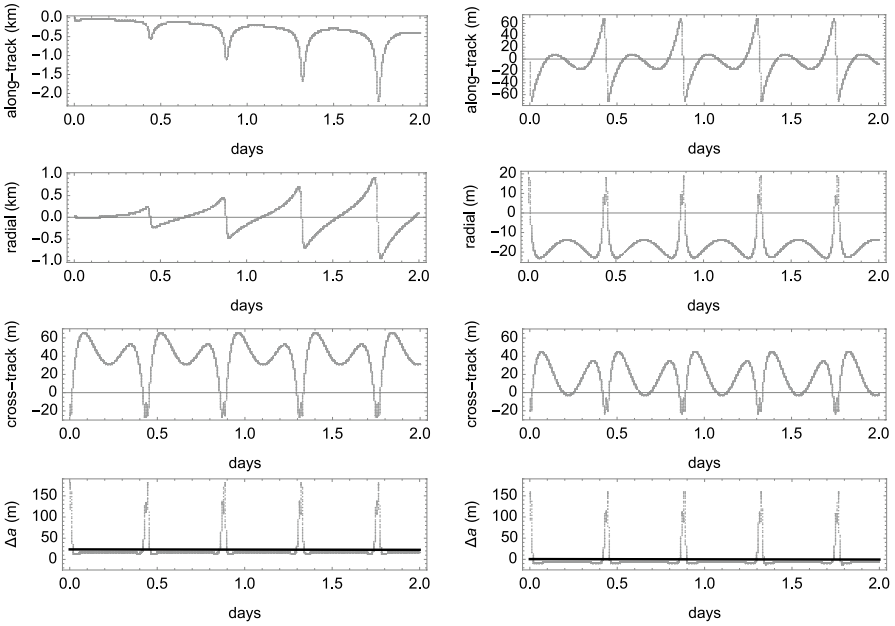

Fig. 4 Errors of a 2-day semianalityc propagation of the GTO test orbit with the 1:2:1 theory (left column) and the properly initialized version of it 1c:2:1 (right column)

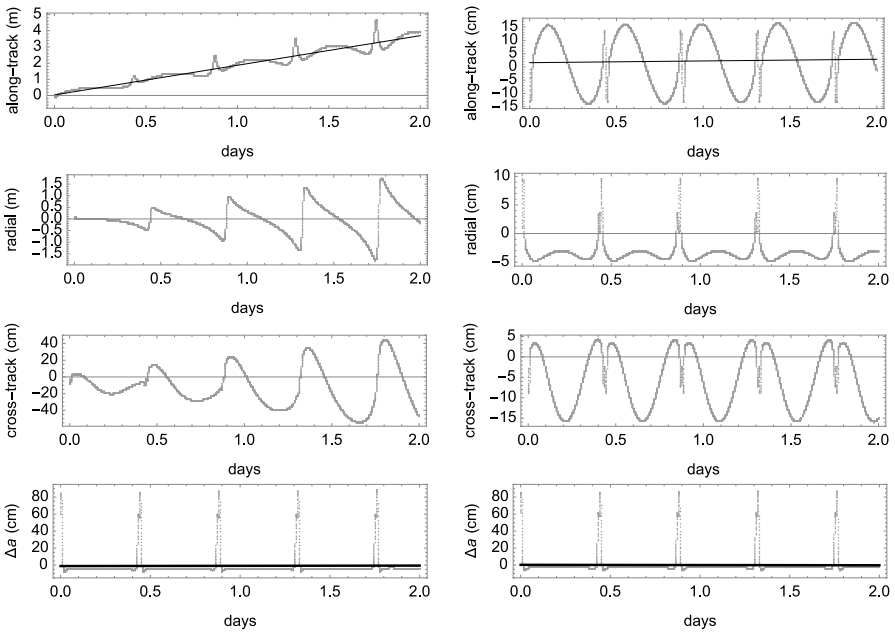

Fig. 5 2-Day semianalityc propagation of the GTO orbit with the 2:2:2 theory (left column) and the calibrated 2c:3:2 theory (right column)

of the mean variations always prevail in the long term. As further illustrated with Fig. 6, this fact could make acceptable to replace the 2:2:2 solution by the much more expedite 1c:2:1 version when position errors of decameters are acceptable in a long-term semianalytical propagation.

It is worth mentioning that the accuracy provided by the semianalytical theory is comparable to the one provided by Brouwer's (correctly initialized) analytical solution for the J_2-problem in analogous time intervals, cf. Fig. 1 of [17]. Still, because Brouwer's solution is limited in application to orbits far enough from the critical inclination, the semianalytical approach may be preferred for it does not suffer an analogous constraint.

6 Conclusion

The derivation of the mean-to-osculating transformation from a generating function is a distinctive feature of the method of Lie transforms. It allowed us to obtain a second-order perturbation solution to the J_2-problem in closed form and arbitrary variables that exactly separates short- and long-period terms. Particularization of the mean-to-osculating transformation for a preferred set of variables only requires fixing the arbitrary integration functions of the slow variables, or "constants," on which this generator depends upon. Our approach takes advantage of the simplicity provided by the set of Delaunay variables in the construction of the perturbation solution. Relying on the traditional set of Keplerian orbital elements, which enjoy analogous characteristics, would have been an acceptable alternative option.

The procedure has been illustrated with the reformulation of the perturbation solution in the set of nonsingular semi-equinoctial elements. To do that it has been imposed that the integration constants of the vectorial generator yield a mean-to-osculating transformation of the semi-equinoctial elements that is purely periodic in the mean distance to the ascending node. An equivalent constraint could have been imposed to a different set of variables, as, for instance, the popular set of equinoctial variables.

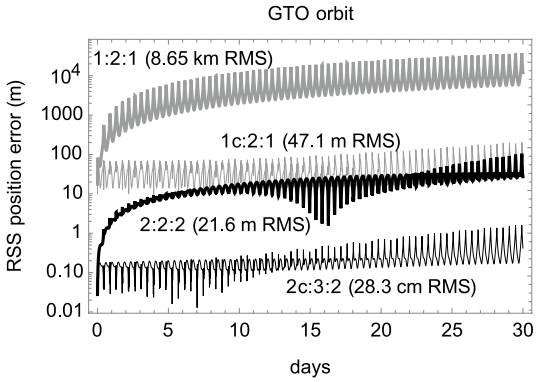

Fig. 6 Position errors of the 1-month propagation of the GTO test orbit with the different versions of the semianalytical theory

The tests presented show that the accuracy of our closed-form theory is neither good nor bad, but the one expected from the truncation order of the series defining the perturbation solution. Failing in achieving a similar accuracy level in comparable time intervals with alternative perturbation solutions simply would mean that the used theory is flawed either in its conception or in its software implementation. Certainly, the finer the programming the higher performance expected from the perturbation solution. But this important issue is not tackled in the paper, where checking that the optimizations provided by the software compiler resulted in semianalytical propagations with the due accuracy seems a more than satisfactory result.

Appendix A: Henrard's Perturbation Theory by Lie Transforms

For two vector functions $\boldsymbol{v}$, $\boldsymbol{w}$, with components $v_j = v_j(\boldsymbol{x})$, $w_j = w_j(\boldsymbol{x})$, $j = 1, 2, \dots n$, of the state variables $\boldsymbol{x} = (x_1, x_2, \dots x_n)$, Henrard [8] defines the differential operator

$$\mathcal{L}_j(v_k) = \sum_{l=1}^{n} \left(\frac{\partial v_j}{\partial x_l} w_l - \frac{\partial w_j}{\partial x_l} v_l \right), \qquad j = 1, 2, \dots n, \tag{A1}$$

which also applies to scalar functions $v_0 = v_0(\boldsymbol{x})$ under the condition $\partial w_0 / \partial x_l = 0$. That is,

$$\mathcal{L}_0(v_0) = \sum_{l=1}^{n} \frac{\partial v_0}{\partial x_l} w_l. \tag{A2}$$

Based on that, Kamel's [7] extension to vectorial flows of Deprit's [6] Lie transforms theory is rebuilt by Henrard as follows.

Let ϵ denote a small parameter, and let

$$W_j = \sum_{m \geq 0} \frac{\epsilon^m}{m!} W_{j,m}(\boldsymbol{x}), \qquad j = 1, 2, \dots n, \tag{A3}$$

be the vectorial generator of the transformation $\boldsymbol{x} = \boldsymbol{x}(\boldsymbol{y}, \epsilon)$ to the new state variables $\boldsymbol{y}$, such that the transformation is obtained as the solution of the differential system

$$\frac{\mathrm{d}x_j}{\mathrm{d}\epsilon} = W_j(\boldsymbol{x}, \epsilon), \qquad j = 1, 2, \dots n, \tag{A4}$$

for the initial conditions $\boldsymbol{x}(\boldsymbol{y}, 0) = \boldsymbol{y}$. A scalar function $F = F(\boldsymbol{x}, \epsilon)$ given by its Taylor series expansion

$$F(\boldsymbol{x}, \epsilon) = \sum_{m \geq 0} \frac{\epsilon^m}{m!} F_{m,0}(\boldsymbol{x}), \qquad F_{m,0} = \left. \frac{\partial^m F}{\partial \epsilon^m} \right|_{\epsilon=0}, \tag{A5}$$

is converted by this transformation into

$$F(\boldsymbol{x}(\boldsymbol{y},\epsilon),\epsilon) = \sum_{m\geq 0} \frac{\epsilon^m}{m!} F_{0,m}(\boldsymbol{y}), \qquad F_{0,m} = \left.\frac{\mathrm{d}^m F}{\mathrm{d}\epsilon^m}\right|_{\epsilon=0}, \tag{A6}$$

through the recursive computation of the coefficients of Deprit's triangle

$$F_{m,q+1} = F_{m+1,q} + \sum_{i=0}^{m} \binom{m}{i} \mathcal{L}_{0,i+1}(F_{m-i,q}), \tag{A7}$$

where $\mathcal{L}_{0,i+1}$ is obtained replacing w_l by $W_{l,i+1}$ in Eq. (A2). The transformation is obtained explicitly from the vectorial generator by first expanding formally

$$x_j = \sum_{m\geq 0} \frac{\epsilon^m}{m!} x_{j,m,0}(\boldsymbol{x}), \qquad j = 1, 2, \ldots n, \tag{A8}$$

where $x_{j,0,0} = x_j$ and $x_{j,m,0} = 0$ for $m > 0$. Then, the coefficients $x_{j,0,m}$ are computed recursively from Deprit's triangle in Eq. (A7), to obtain

$$x_j = \sum_{m\geq 0} \frac{\epsilon^m}{m!} x_{j,0,m}(\boldsymbol{y}). \tag{A9}$$

The vectorial generator in Eq. (A3) is analogously used for transforming the perturbation flow

$$\frac{\mathrm{d}x_j}{\mathrm{d}t} = X_j(\boldsymbol{x},\epsilon) = \sum_{m\geq 0} \frac{\epsilon^m}{m!} \Phi_{j,m,0}(\boldsymbol{x}). \tag{A10}$$

into

$$\frac{\mathrm{d}y_j}{\mathrm{d}t} = \sum_{k\geq 1}^{n} \frac{\partial y_j}{\partial x_k} X_j(\boldsymbol{x},\epsilon) = Y_j(\boldsymbol{x},\epsilon), \tag{A11}$$

where, in the new variables

$$Y_j(\boldsymbol{x}(\boldsymbol{y},\epsilon),\epsilon) = \sum_{m\geq 0} \frac{\epsilon^m}{m!} \Phi_{j,0,m}(\boldsymbol{y}). \tag{A12}$$

The new coefficients $\Phi_{j,0,m}$ are obtained from the recursion

$$\Phi_{j,m,q+1} = \Phi_{j,m+1,q} + \sum_{i=0}^{m} \binom{m}{i} \mathcal{L}_{j,i+1}(\Phi_{k,m-i,q}), \qquad j,k = 1, 2, \ldots, n \tag{A13}$$

where $\mathcal{L}_{j,i+1}$ is now computed from Eq. (A1) in which w_j and w_l are replaced by $W_{j,i+1}$ and $W_{l,i+1}$, respectively.

In a perturbation theory, at each step m of the perturbation approach, the desired form of the coefficients $\Phi_{j,0,m}$ is chosen, and those $W_{j,i+1}$ are solved from the homological equation

$$\mathcal{L}_{j,m}(\Phi_{k,0,0}) = \Phi_{j,0,m} - \tilde{\Phi}_{j,0,m}, \quad k, j = 1, 2, \dots n. \tag{A14}$$

According to Eq. (A1), Eq. (A14) is a partial differential system in the coefficients $W_{j,m}$, in which $\tilde{\Phi}_{j,0,m}$ denotes the computable terms obtained at the step m of the perturbation approach from repeated application of Deprit's triangle in the variant of Eq. (A13).

Perturbed Kepler motions are commonly expressed in terms of orbital elements, with five elements of the constant type and one element of the linear type. The variation of the latter—either the mean anomaly or a related fast-evolving angle—is the only one affected of a zeroth-order term in the perturbation arrangement, who besides only depends on a single variable that is related to the semimajor axis. In consequence, it is assumed that $\Phi_{1,0,0} = \Phi_{1,0,0}(x_4)$ and $\Phi_{k,0,0} = 0$ otherwise. Therefore, according to Eqs. (A1) and (A14),

$$W_{j,m} = \frac{1}{\Phi_{1,0,0}} \int \left(\tilde{\Phi}_{j,0,m} - \Phi_{j,0,m}\right) \mathrm{d}x_1, \qquad j = 2, \dots 6, \tag{A15}$$

$$W_{1,m} = \frac{1}{\Phi_{1,0,0}} \int \left(\tilde{\Phi}_{1,0,m} - \Phi_{1,0,m} + \frac{\partial \Phi_{1,0,0}}{\partial x_4} W_{4,m}\right) \mathrm{d}x_1. \tag{A16}$$

The inverse transformation is needed to properly initialize the integration of Eq. (A11). A convenient feature of the method of Lie transforms is that the generator of the inverse transformation V_j is just the opposite of the direct one in Eq. (A3). According to Eq. (A4), W_j is a vectorial flow, and hence $V_j = -W_j(\boldsymbol{x}(\boldsymbol{y}, \epsilon), \epsilon)$ is directly formulated in the $\boldsymbol{y}$ variables by the straightforward application of the recursion in Eq. (A13) to $-W_j(\boldsymbol{x}, \epsilon)$. Once the generator is computed, the inverse transformation is obtained from analogous equations to Eqs. (A8)–(A9), in which now the scalar differential operator $\mathcal{L}_{0,i+1}$ used in Deprit's recursion (A7) is obtained replacing w_l by $V_{l,i+1}$ in Eq. (A2).

Appendix B: First Order Generating Function in Delaunay Variables

For completeness, the first-order terms of the vectorial generating function are borrowed from [29, 30], where first indices $1, 2, \dots 6$ correspond to the Delaunay variables ℓ, g, h, L, G, and H, respectively, and the second index 1 denotes first order of J_2. Thus,

$$\begin{aligned} W_{1,1} = \frac{R_\oplus^2}{p^2} \frac{\eta}{32e} \Big\{ & (3s^2 - 2)\left[2e^2 \sin 3f - 6(e^2 - 4) \sin f + 12e \sin 2f\right] + s^2 \left[3e^2 \right. \\ & \times \sin(2\omega - f) - 2\frac{4\eta^3 - \eta^2 - 18\eta - 9}{(1+\eta)^2} e \sin 2\omega + 3(5e^2 + 4) \sin(2\omega + f) \\ & \left. + (e^2 - 28) \sin(2\omega + 3f) - 18e \sin(2\omega + 4f) - 3e^2 \sin(2\omega + 5f)\right] \Big\}, \end{aligned} \tag{B17}$$

$$
\begin{aligned}
W_{2,1} = &-\frac{R_\oplus^2}{p^2}\frac{3}{4}(5s^2-4)\phi + \frac{R_\oplus^2}{p^2}\frac{1}{32e}\Big\{-\left[6e^2(17s^2-14)+24(3s^2-2)\right]\sin f \\
&-12e(3s^2-2)\sin 2f - 2e^2(3s^2-2)\sin 3f - 3e^2s^2\sin(2\omega - f) \\
&-\frac{2e}{(1+\eta)^2}\left[\eta^3(16s^2-8)+\eta^2(11s^2-4)+(1+2\eta)(4-s^2)\right]\sin 2\omega \\
&+\left[3e^2(15s^2-8)-12s^2\right]\sin(2\omega+f) + 12e(5s^2-2)\sin(2\omega+2f) \\
&+\left[e^2(19s^2-8)+28s^2\right]\sin(2\omega+3f) + 18es^2\sin(2\omega+4f) \\
&+3e^2s^2\sin(2\omega+5f)\Big\},
\end{aligned}
\tag{B18}
$$

$$
\begin{aligned}
W_{3,1} = &-\frac{R_\oplus^2}{p^2}\frac{3}{2}c\phi + \frac{R_\oplus^2}{p^2}\frac{1}{4}c\Big\{-6e\sin f + \frac{1+2\eta}{(1+\eta)^2}e^2\sin 2\omega + 3e\sin(2\omega+f) \\
&+3\sin(2\omega+2f) + e\sin(2\omega+3f)\Big\},
\end{aligned}
\tag{B19}
$$

$$
\begin{aligned}
W_{4,1} = L\frac{R_\oplus^2}{p^2}\frac{1}{32\eta^2}\Big\{&(3s^2-2)\big[4(\eta-1)(2\eta^2+5\eta+5)-6(e^2+4)e\cos f - 12e^2 \\
&\times\cos 2f - 2e^3\cos 3f\big] + s^2\Big[3e^3\cos(2\omega-f) + 18e^2\cos 2\omega + 9(e^2+4) \\
&\times e\cos(2\omega+f) + 12(3e^2+2)\cos(2\omega+2f) + 9(e^2+4)e\cos(2\omega+3f) \\
&+18e^2\cos(2\omega+4f) + 3e^3\cos(2\omega+5f)\Big]\Big\}.
\end{aligned}
\tag{B20}
$$

$$
\begin{aligned}
W_{5,1} = &G\frac{R_\oplus^2}{p^2}\frac{1}{4}s^2\Big[e\cos(2\omega+3f) + 3\cos(2\omega+2f) + 3e\cos(2\omega+f) \\
&+\frac{1+\eta-2\eta^2}{1+\eta}\cos 2\omega\Big],
\end{aligned}
\tag{B21}
$$

in which $\phi = f - \ell$ is the equation of the center.

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Declarations

Conflict of Interest The author declares no conflict of interest.

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